PH-511: Mathematical Physics - Project Report

The Confluent Hypergeometric Functions

Theory, Computation and Applications

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Abstract

The confluent hypergeometric function $M(a,b,z) = {}_1F_1(a;b;z)$ solves Kummer's differential equation and contains, as special cases, many of the functions used in physics and engineering: exponentials, the error function, Bessel functions, Hermite and Laguerre polynomials and the radial wave functions of the hydrogen atom. In this report we define the function, derive its basic properties, and verify them with the software tools SymPy, mpmath and SciPy. We plot the function for several parameter values, reproduce hydrogen-atom radial wave functions from it, and summarise some areas of current research in which it appears.

1 Introduction

Many of the second-order linear differential equations of mathematical physics have the form

$$z\,w''(z) + (b - z)\,w'(z) - a\,w(z) = 0, \tag{1}$$

known as *Kummer's equation* or the confluent hypergeometric equation. It has a regular singular point at $z = 0$ and an irregular singular point at $z = \infty$. It is obtained from the Gauss hypergeometric equation, which has three regular singular points at 0, 1, ∞, by letting the singularity at 1 merge ("*confluence*") with the one at ∞. This is the origin of the name *confluent* hypergeometric equation [1, 4].

The solution regular at $z = 0$ is Kummer's function $M(a,b,z)$, also written ${}_1F_1(a;b;z)$ or $\Phi(a,b,z)$. A second, independent solution is Tricomi's function $U(a,b,z)$. These functions were studied by Kummer in 1836 and appear in a surprisingly wide range of problems: the quantum-mechanical hydrogen atom, the Coulomb scattering problem, the harmonic oscillator, probability distributions, and signal-processing models [2, 3].

The aim of this project is to (i) understand the theory of $M(a,b,z)$, (ii) implement and verify its main properties using computer software, and (iii) see where it is used in present-day research. Section 2 develops the theory, Section 3 describes the software implementation and plots, Section 4 discusses applications, Section 5 concludes, and Sections 6 and 7 list the roles of the team members and acknowledgements.

2 Theory

2.1 Definition and series solution

We look for a solution of Eq. (1) by the Frobenius method, $w = \sum_{n\ge 0} c_n z^{n+s}$. The indicial equation is $s(s + b - 1) = 0$, so $s = 0$ or $s = 1 - b$. For $s = 0$ the recursion relation is

$$c_{n+1} = \frac{(a+n)}{(b+n)(n+1)}\,c_n,$$

which gives, with $c_0 = 1$,

$$M(a,b,z) = \sum_{n=0}^{\infty} \frac{(a)_n}{(b)_n}\frac{z^n}{n!} = 1 + \frac{a}{b}z + \frac{a(a+1)}{b(b+1)}\frac{z^2}{2!} + \cdots, \tag{2}$$

where $(a)_n = a(a+1)\cdots(a+n-1) = \Gamma(a+n)/\Gamma(a)$ is the Pochhammer symbol. By the ratio test, the series converges for all finite z, so $M(a,b,z)$ is an *entire* function of z (provided $b \neq 0, -1, -2, \ldots$). If $a = -m$ is a non-positive integer the series terminates and M becomes a polynomial of degree m.

For $s = 1 - b$ one finds the second solution $z^{1-b}M(a-b+1, 2-b, z)$ (when b is not an integer). A convenient linear combination which is well defined for all b is Tricomi's function

$$U(a,b,z) = \frac{\Gamma(1-b)}{\Gamma(a-b+1)}M(a,b,z) + \frac{\Gamma(b-1)}{\Gamma(a)}z^{1-b}M(a-b+1, 2-b, z). \tag{3}$$

2.2 Integral representations

For $\operatorname{Re} b > \operatorname{Re} a > 0$, expanding e^{zt} in a power series and using the Beta integral gives Euler's integral

$$M(a,b,z) = \frac{\Gamma(b)}{\Gamma(a)\Gamma(b-a)}\int_0^1 e^{zt}\,t^{a-1}(1-t)^{b-a-1}\,dt. \tag{4}$$

Similarly, for $\operatorname{Re} a > 0$, $\operatorname{Re} z > 0$,

$$U(a,b,z) = \frac{1}{\Gamma(a)} \int_0^\infty e^{-zt}\, t^{a-1}(1+t)^{b-a-1}\, dt. \tag{5}$$

These formulas show the link with the Gamma and Beta functions, and are convenient for numerical evaluation and for finding asymptotics.

2.3 Basic identities

The following identities can be proved directly from the series (2) or from Eq. (4) (see [2, 4], Chapter 13):

$$\text{Kummer's transformation:} \quad M(a,b,z) = e^z\, M(b-a,b,-z), \tag{6}$$

$$\text{Derivative:} \quad \frac{d}{dz} M(a,b,z) = \frac{a}{b}\, M(a+1,b+1,z), \tag{7}$$

$$\text{Contiguous relation:} \quad b\, M(a,b,z) - b\, M(a-1,b,z) - z\, M(a,b+1,z) = 0. \tag{8}$$

The transformation (6) is very useful: for $z \to -\infty$ it turns the behaviour of M into that at $+\infty$.

2.4 Asymptotic behaviour

For large $|z|$ one has [2, 4]

$$M(a,b,z) \sim \frac{\Gamma(b)}{\Gamma(a)}\, e^z z^{a-b} \quad (z \to +\infty), \qquad M(a,b,z) \sim \frac{\Gamma(b)}{\Gamma(b-a)}\, (-z)^{-a} \quad (z \to -\infty). \tag{9}$$

Thus M grows exponentially along the positive real axis (unless a is a non-positive integer, when it is a polynomial) and decays like a power law along the negative real axis. This difference is what produces *quantisation* conditions in physical problems (see Section 2.6).

2.5 Special cases

Many familiar functions are particular cases of $M(a,b,z)$:

Function	In terms of M
Exponential	$M(a,a,z) = e^z$
Elementary	$M(1,2,z) = (e^z - 1)/z$
Error function	$\operatorname{erf}(z) = \dfrac{2z}{\sqrt{\pi}}\, M\left(\frac{1}{2}, \frac{3}{2}, -z^2\right)$
Laguerre polynomial	$L_n^{(\alpha)}(z) = \dbinom{n+\alpha}{n} M(-n, \alpha+1, z)$
Hermite polynomial	$H_{2n}(x) = (-1)^n \dfrac{(2n)!}{n!}\, M\left(-n, \frac{1}{2}, x^2\right)$
Bessel function	$J_\nu(z) = \dfrac{(z/2)^\nu e^{-iz}}{\Gamma(\nu+1)}\, M\left(\nu+\frac{1}{2}, 2\nu+1, 2iz\right)$

In this sense $M(a,b,z)$ is a "master function" from which many special functions are derived.

2.6 Relation with the hydrogen atom

The radial Schrödinger equation for the hydrogen atom, with $u = rR$ and in atomic units ($\hbar = m = e^2/4\pi\epsilon_0 = 1$), is

$$u'' + \left[\frac{2}{r} - \frac{l(l+1)}{r^2} + 2E\right] u = 0.$$

Writing $E = -1/(2\nu^2)$, $\rho = 2r/\nu$ and $u = \rho^{l+1} e^{-\rho/2} v(\rho)$ reduces this to Kummer's equation (1) with $a = l + 1 - \nu$ and $b = 2l + 2$. The solution regular at the origin is $v = M(l+1-\nu, 2l+2, \rho)$. By Eq. (9) this grows like e^ρ and makes the wave function non-normalisable, *unless the series terminates*, i.e. $l + 1 - \nu = -n_r$ with $n_r = 0, 1, 2, \ldots$. Hence $\nu = n = n_r + l + 1$ is an integer and

$$E_n = -\frac{1}{2n^2}, \qquad R_{nl}(r) \propto r^l e^{-r/n}\, M\left(-n+l+1,\, 2l+2,\, \frac{2r}{n}\right), \tag{10}$$

which is the familiar Bohr spectrum together with the Laguerre-type radial functions.

3 Worked examples

3.1 Example 1: a simple polynomial solution

Choose $a = -1$ and $b = 2$. From the series,

$$M(-1, 2, z) = 1 + \frac{-1}{2}z + \underbrace{\frac{(-1)_2}{(2)_2}\frac{z^2}{2!}}_{=0} + \cdots .$$

Since $(-1)_2 = (-1)(0) = 0$, all terms after the linear term vanish. Thus

$$\boxed{M(-1, 2, z) = 1 - \frac{z}{2}.}$$

Direct verification is possible. Here $y = 1 - z/2$, so $y' = -1/2$ and $y'' = 0$. Substitution into Kummer's equation gives

$$z(0) + (2 - z)(-1/2) - (-1)(1 - z/2) = 0.$$

Therefore the expression is indeed a solution.

3.2 Example 2: exponential special case

If $a = b$ and b is admissible, then $(a)_n/(b)_n = 1$. Hence

$$M(a, a, z) = \sum_{n=0}^{\infty} \frac{z^n}{n!} = e^z.$$

For example, $M(1, 1, z) = e^z$. This is a useful check for software calculations.

3.3 Example 3: first terms for $M(1, 2, z)$

For $a = 1, b = 2$,

$$M(1, 2, z) = 1 + \frac{1}{2}z + \frac{1}{3}\frac{z^2}{2!} + \frac{1}{4}\frac{z^3}{3!} + \cdots ,$$

so

$$M(1, 2, z) = 1 + \frac{z}{2} + \frac{z^2}{6} + \frac{z^3}{24} + \cdots .$$

This series is useful for numerical approximation when z is modest. It also has the closed form $M(1, 2, z) = (e^z - 1)/z$ for $z \neq 0$, with the continuous value 1 at $z = 0$. Expanding the exponential confirms the series above.

4 Implementation in a Software

We used Python with the libraries SymPy (symbolic computation) [5], mpmath (arbitrary-precision numerics) [6] and SciPy/Matplotlib (fast numerics and plotting) [7]. The same checks can be done in Mathematica/Wolfram Alpha using `Hypergeometric1F1[a,b,z]`, or in Matlab using `kummerU` and `hypergeom([a],[b],z)`.

4.1 Verification of the differential equation

The truncated series (2) (up to z^8) was substituted symbolically into Kummer's equation. The only surviving term is the one arising from truncation:

$$zM'' + (b - z)M' - aM = -\frac{(a)_9}{(b)_8\, 8!}\, z^8,$$

which is exactly what is expected: every coefficient up to z^7 cancels, confirming the recursion relation. SymPy's `hyperexpand` also reproduces the closed forms $M(a, a, z) = e^z$ and $M(1, 2, z) = (e^z - 1)/z$.

Identity tested	Values compared	Difference
Kummer transformation (6)	2.025023369966… both sides	0
Euler integral (4)	2.025023369966… both sides	1.7×10^{-24}
ODE residual (numerical derivatives)	$zM'' + (b-z)M' - aM$	2.0×10^{-31}
Laguerre, $n = 4, \alpha = 1.5, x = 2.2$	$-1.576162\ldots$ both sides	$< 10^{-15}$
Error function, $x = 1.3$	$0.934007944\ldots$ both sides	$< 10^{-15}$

Table 1: Numerical checks of the identities of Section 2.

4.2 Numerical verification of identities

Table 1 summarises the numerical tests carried out with 30-digit precision (MPMATH) for the test values $a = 0.7$, $b = 2.3$, $z = 1.9$. For the asymptotic formula (9) with $z = 30$ the exact value of $M(0.7, 2.3, 30)$ is 4.2309×10^{10} while the leading asymptotic term gives 4.1602×10^{10}, a relative difference of about 1.7%, consistent with the $O(1/z)$ correction.

4.3 Numerical plots and simulation results

The following figures present the main computational results. They are grouped by the mathematical or physical question they address: numerical integration of Kummer's equation, independent evaluation through Euler's integral and the series, the hydrogen-atom application, and the asymptotic limits.

4.3.1 Numerical solution of Kummer's differential equation

Figure 1 compares the numerical solution obtained by integrating Kummer's differential equation with the analytical hypergeometric function. The agreement provides a direct check of the ODE implementation.

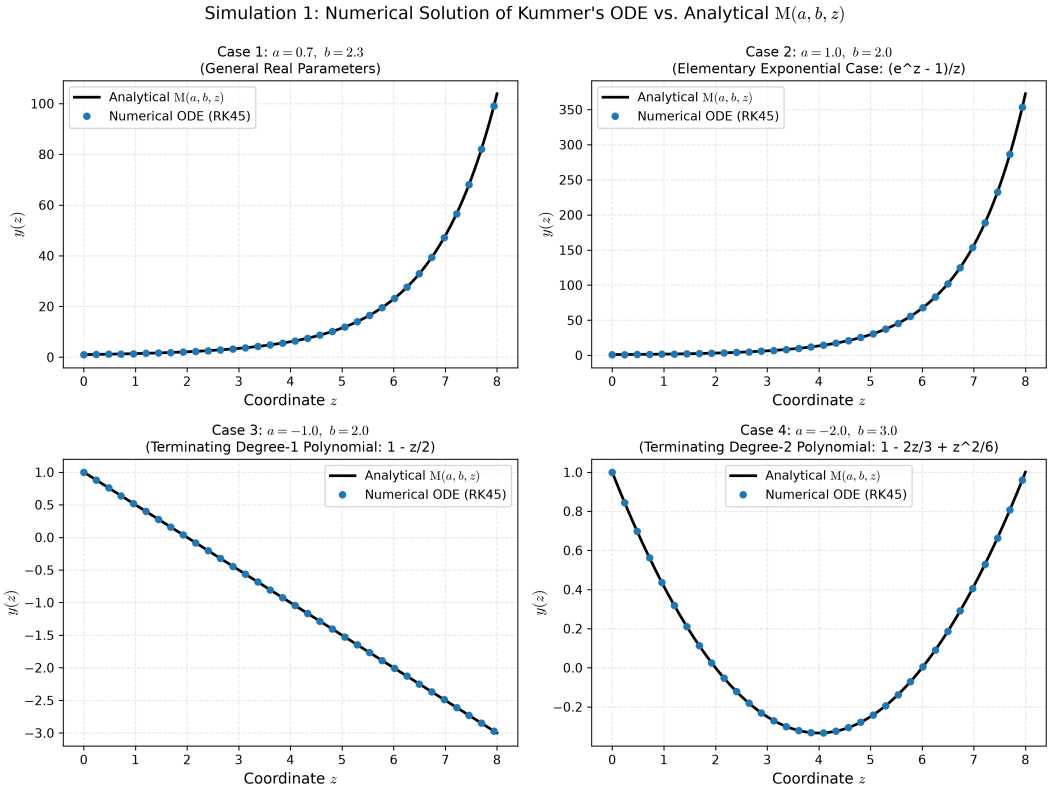


Figure 1: Analytical solution $M(a, b, z)$ compared with the numerical RK45 solution of Kummer's differential equation for representative parameter sets.

The absolute-error plot in Fig. 2 examines numerical accuracy for several solver tolerances. It illustrates how tightening the relative and absolute error controls reduces the discrepancy between the

numerical trajectory and the analytical reference.

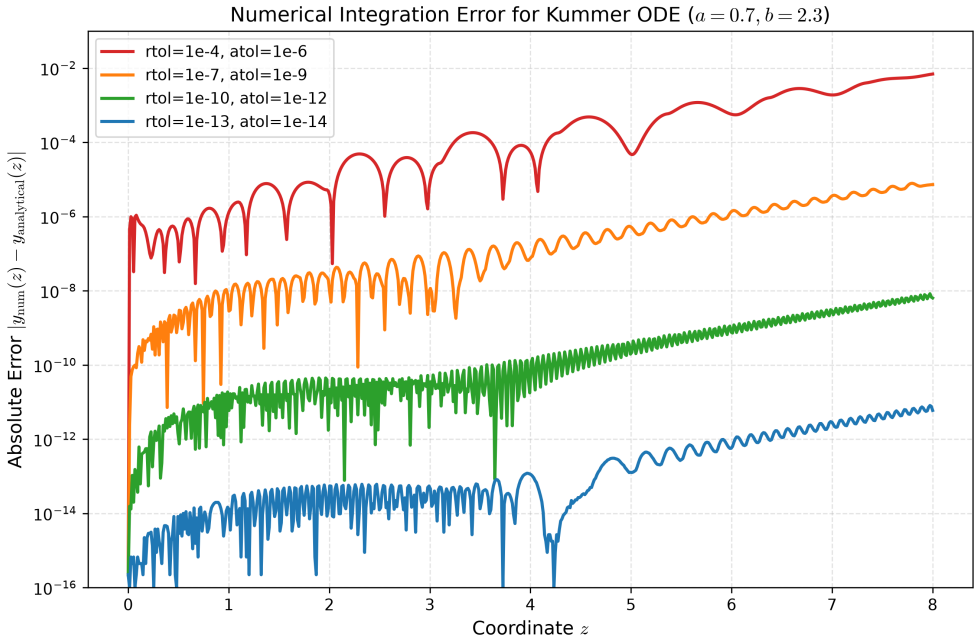


Figure 2: Absolute error of the numerical ODE solution for different solver tolerances.

4.3.2 Euler integral representation and series convergence

For parameters satisfying $\operatorname{Re} b > \operatorname{Re} a > 0$, Euler's integral provides an independent numerical route to $M(a, b, z)$. Figure 3 compares the integral quadrature with the manually implemented power series and the SciPy reference implementation. Agreement among the three approaches is a useful cross-check because they use different computational procedures.

Figure 4 shows how the truncated Taylor–Pochhammer series approaches the reference value as more terms are included. The convergence rate depends on z: larger positive arguments generally require more terms before the truncation error becomes small.

4.3.3 Hydrogen-atom radial wave functions and energy levels

The hydrogen-atom application connects the mathematical termination condition of Kummer's series to physically acceptable bound states. Figure 5 displays radial wave functions for different quantum states. Their zeros away from the origin are radial nodes; the expected number is $n - l - 1$.

The radial probability distribution is $P_{nl}(r) = r^2 |R_{nl}(r)|^2$. Unlike the signed radial wave function, this quantity is non-negative and describes the probability density with respect to radial distance, so that $P_{nl}(r)\, dr$ is the probability of finding the electron in a spherical shell between r and $r + dr$.

The bound-state energy depends on the principal quantum number n and is given, in atomic units, by $E_n = -1/(2n^2)$. Figure 7 illustrates the resulting discrete spectrum. The requirement that the wave function remain normalisable forces the confluent hypergeometric series to terminate, producing the allowed integer values of n.

4.3.4 Positive- and negative-argument asymptotics

The asymptotic behaviour of Kummer's function is different on the positive and negative real axes. For large positive z, the dominant contribution grows exponentially, whereas for large negative z the leading behaviour is algebraic. Figure 8 compares the function with these respective leading asymptotic forms and makes this directional difference visible.

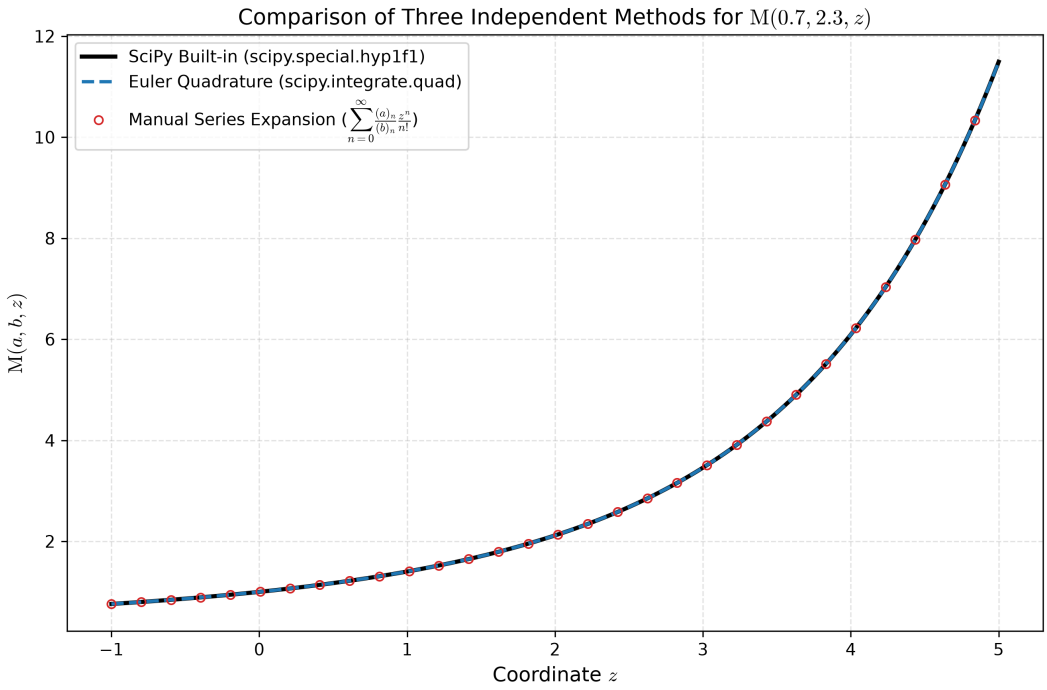


Figure 3: Comparison of Kummer's function evaluated using Euler integral quadrature, a manually implemented series recurrence, and SciPy.

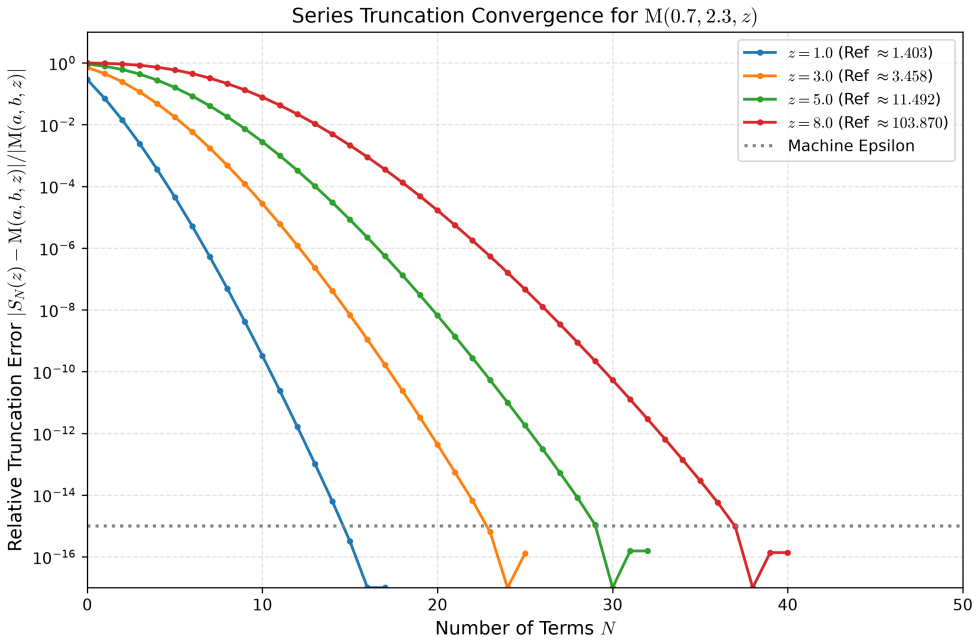


Figure 4: Convergence of the manually evaluated power series as a function of the number of retained terms for several values of z.

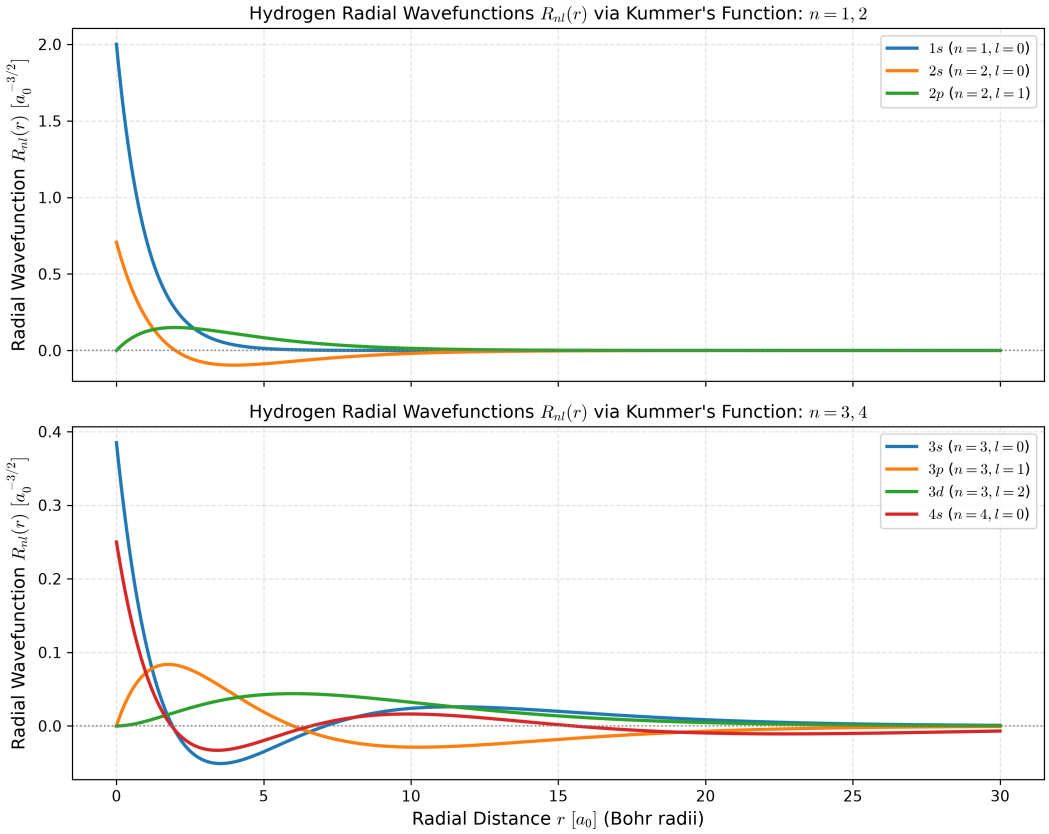


Figure 5: Hydrogen-atom radial wave functions $R_{nl}(r)$ for representative quantum states.

4.4 Remarks on numerical computation

The series (2) is fine for moderate $|z|$, but for large negative z it suffers from cancellation between large alternating terms. Software libraries therefore switch between the series, Kummer's transformation (6) and asymptotic expansions depending on the arguments. We observed that using Kummer's transformation for $z < 0$ is more stable, and that arbitrary-precision arithmetic (MPMATH) is the safest way to cross-check results from SciPy. As the instructions suggest, results obtained from AI chatbots or any single package should be verified independently; we did this by comparing the integral representation, the series and two different software packages.

5 Application in Current Research

Because $M(a, b, z)$ contains so many special functions, it appears in diverse areas. We summarise a few of them. (Specific recent papers found on arXiv / Google Scholar / Inspire HEP should be added to the bibliography; see the note at the end of this section.)

5.1 Quantum mechanics and atomic physics.

The Coulomb problem (bound states and scattering) is solved with M and U; the Coulomb wave functions that describe charged-particle scattering in nuclear and atomic physics are of Kummer type [1, 2]. The same function appears for the 3D harmonic oscillator, the Morse potential, and various exactly solvable potentials used as test cases in current work on quantum systems.

5.2 Wireless communications and statistics.

The Nakagami-m fading model, the non-central chi-square distribution and the Rician distribution involve $M(a, b, z)$ in their moment generating functions or probability densities. Performance analysis (error probabilities, outage capacity) of modern wireless systems is therefore often expressed using ${}_1F_1$ [8].

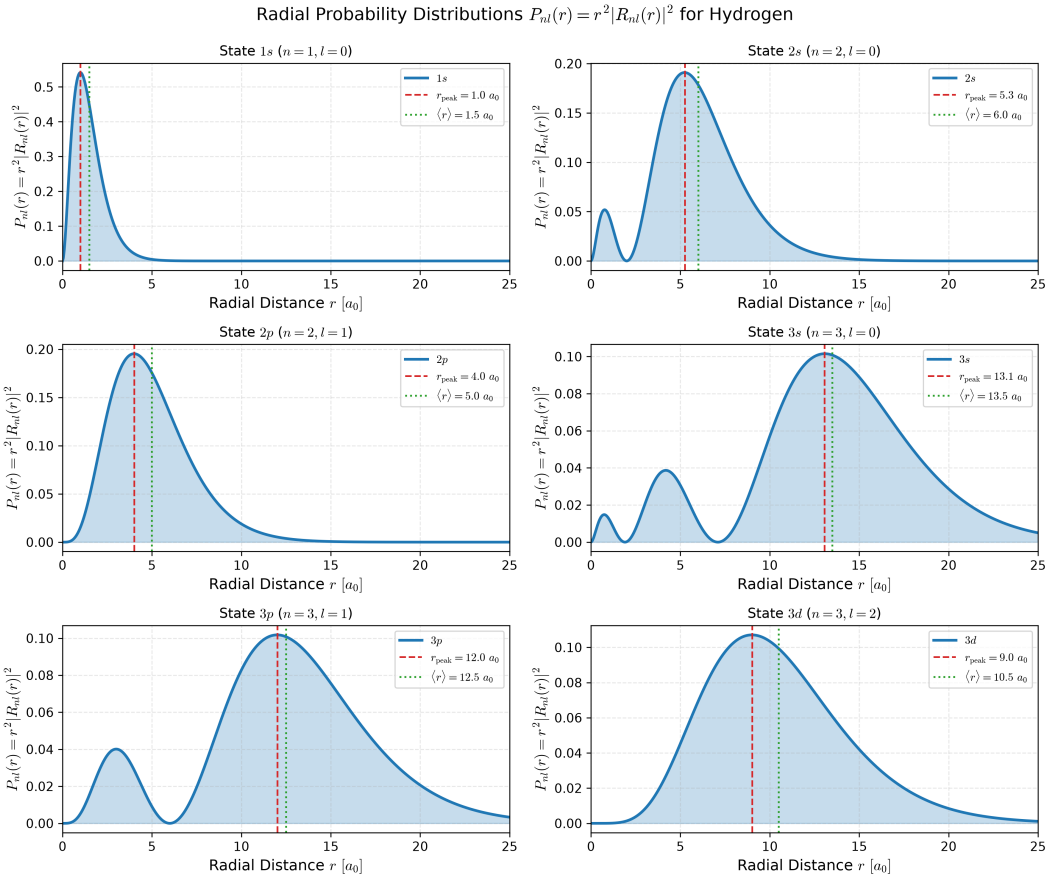


Figure 6: Radial probability distributions $P_{nl}(r) = r^2|R_{nl}(r)|^2$ for representative hydrogen states.

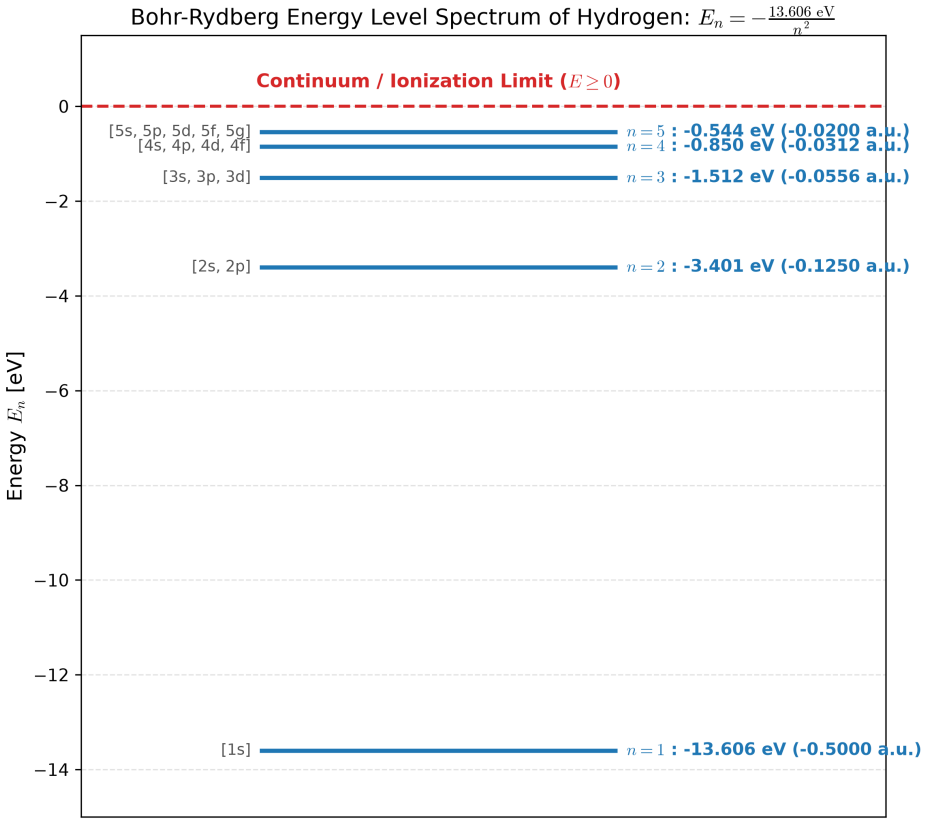


Figure 7: Discrete hydrogen-atom energy levels for different principal quantum numbers.

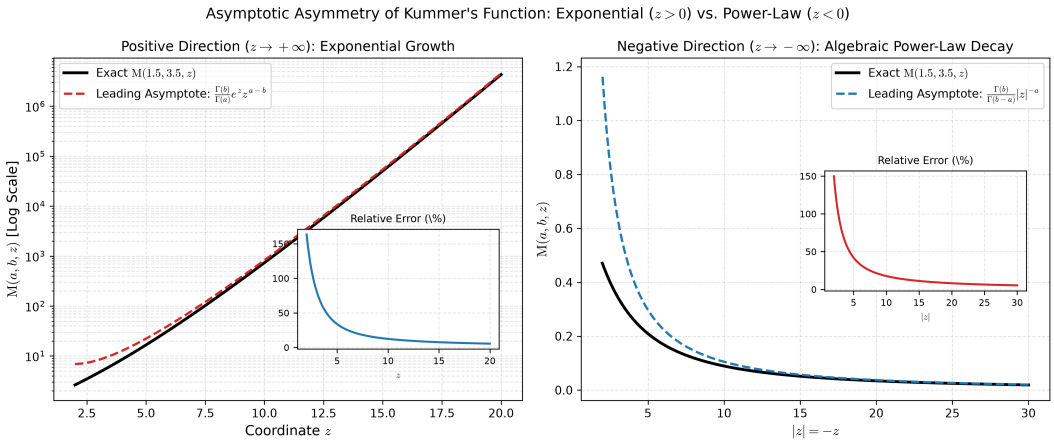


Figure 8: Positive- and negative-argument asymptotic behaviour of Kummer's function, showing exponential growth for large positive z and algebraic behaviour for large negative z.

5.3 Gravitational and field-theoretic problems.

Perturbation equations on black-hole and other curved backgrounds frequently reduce to confluent hypergeometric (or its generalisation, the confluent Heun) equations; the connection between the solution near the horizon and at infinity is done with the asymptotic formulas (9). Likewise, particle production in strong electric fields (Schwinger effect) and in expanding universes is described with Whittaker/Kummer functions.

5.4 Mathematical methods.

The Kummer function serves as a generating building block for generalised hypergeometric functions ${}_pF_q$ used in Feynman-integral calculations, in random-matrix theory, and in the theory of orthogonal polynomials.

6 Conclusion

We studied the confluent hypergeometric function $M(a, b, z)$ from its definition as the regular solution of Kummer's equation. We derived the series, the Euler integral representation, the main identities and the asymptotic behaviour, and showed how many well-known special functions are special cases. We verified the theory using SymPy, mpmath and SciPy, produced plots for different parameters, and used the function to reproduce the hydrogen-atom spectrum and radial wave functions. The key physical lesson is that termination of the series ($a = -n_r$) is equivalent to the quantisation condition for the bound states. Finally, we saw that the function remains relevant in modern research in physics, statistics and communications.

7 Roles and Responsibilities

Team member	Contribution
Aman Kumar	Literature review and theory
Md Farhan	Introduction, theory (series, integral representation)
Abhilasha	Software implementation, verification and plots
Arani Ghosh	Applications in current research, bibliography, LaTeX typesetting
Shashank	Reading, discussion and proof-reading of the report

8 Acknowledgements

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